COURTADE'S VOLUME CONJECTURE FOR MINKOWSKI SUMS IS FALSE

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Abstract. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: Courtade's geometric analogue of entropy-power concavity: $|K+L+B|^{1/d}|B|^{1/d} + |K|^{1/d}|L|^{1/d} \le |K+B|^{1/d}|L+B|^{1/d}$ for bounded convex sets. *Certificate:* A zonotope with seven integer generators and two segments; four integer volumes with $68 \cdot 44 = 2992 < 3024 = 28 \cdot 108$.

Posed by Thomas A. Courtade (2017); counterexample found by Suvrit Sra, by hand (2022-05); formalized by Claude Fable 5; audited by Claude Fable 5; contributed by Suvrit Sra.

For two sets $K, L \subset \mathbb{R}^d$ the Minkowski sum is the set $K + L := \{x + y : x \in K,\ y \in L\}$. The statement below is the closing question of Courtade (2017); the "Theorem 1" refers to Courtade's similarly structured entropy-power inequality, of which he conjectured a volume analogue. Dropping $V(K)^{1/d}V(L)^{1/d}$ weakens his question to the *log-submodularity* of volume under Minkowski summation, $V(B)V(K + L + B) \le V(K + B)V(L + B)$, a property with its own literature; see the remarks after the proof. Following Courtade, we'll locally also use the notation $|K| \equiv V(K)$.

Problem 1 (As posed: Closing remarks (Section IV) of Courtade (2017), posed as a question for future work; no problem number is assigned there.)**.** In particular, for K, L, B bounded convex sets in $\mathbb{R}^d$, does it hold that:

$$|K + L + B|^{1/d}|B|^{1/d} + |K|^{1/d}|L|^{1/d} \le |K + B|^{1/d}|L + B|^{1/d},$$

where $|\cdot|$ denotes d-dimensional volume?

Problem 1 asks for the geometric analogue of the entropy-power inequality that Courtade (2017) proves for Gaussian perturbations: volumes of Minkowski sums should play the role of entropy powers of independent sums. For $d = 1$ the two sides agree identically, and for $d = 2$ the inequality is true (Fradelizi et al., 2024a). It fails from dimension three onwards. We give a fully explicit witness in the zonotope calculus, where every volume in sight is an integer.

Theorem 2 ([**Disproof**]: Courtade's volume conjecture)**.** *For every $d \ge 4$ there are bounded convex sets $K, L, B \subset \mathbb{R}^d$ with*

$$|K + L + B|^{1/d}|B|^{1/d} + |K|^{1/d}|L|^{1/d} > |K + B|^{1/d}|L + B|^{1/d}.$$

One may take B a zonotope and K, L segments; in $\mathbb{R}^4$, B has seven integer generators and all three sets may moreover be taken full-dimensional.

Proof. A zonotope $Z(v_1, \dots, v_m) = \sum_{i \le m}[0, v_i] \subset \mathbb{R}^d$ has volume $\sum_S |\det(v_S)|$, the sum running over all d-element subsets S of the generators (Shephard, 1974), and Minkowski summation simply

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

concatenates generator lists. In $\mathbb{R}^4$ let B be the zonotope with the seven columns of

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

as generators, and let $K = [0, b]$ and $L = [0, c]$ be the segments with $b = (1, -1, 1, 1)$ and $c = (0, 0, -1, 1)$. Summing $|\det|$ over the $\binom{7}{4} = 35$ generator quadruples of B (eleven are singular, twenty have $|\det| = 1$, four have $|\det| = 2$) gives $|B| = 28$, and likewise over the $\binom{8}{4} = 70$ quadruples of $B + K$ and of $B + L$ and the $\binom{9}{4} = 126$ of $B + K + L$:

$$|B| = 28, \qquad |K + B| = 68, \qquad |L + B| = 44, \qquad |K + L + B| = 108.$$

Since $|K| = |L| = 0$, the conjectured inequality asserts $(28 \cdot 108)^{1/4} \le (68 \cdot 44)^{1/4}$. Raising to the fourth power, which is monotone on nonnegative reals, this is $3024 \le 2992$ — false.

For full-dimensional sets in $\mathbb{R}^4$, thicken each segment to a box: with $\varepsilon = \frac{1}{100}$ replace K by $K_\varepsilon = [0, b] + [0, \varepsilon]^4$ and L by $L_\varepsilon = [0, c] + [0, \varepsilon]^4$, both zonotopes with five generators and positive volume ($|K_\varepsilon| = \frac{401}{10^8}$, $|L_\varepsilon| = \frac{201}{10^8}$). The same generator sums give the exact rationals

$$|K_\varepsilon + B| = \tfrac{6926731601}{10^8}, \qquad |L_\varepsilon + B| = \tfrac{4484551401}{10^8}, \qquad |K_\varepsilon + L_\varepsilon + B| = \tfrac{696978401}{6250000},$$

whence $|B|\,|K_\varepsilon + L_\varepsilon + B| - |K_\varepsilon + B|\,|L_\varepsilon + B| = \frac{1613484591844769 99}{10^{16}} > 0$. In fourth powers the conjectured inequality fails a fortiori, its left side now carrying the additional positive term $|K_\varepsilon|^{1/4}|L_\varepsilon|^{1/4}$.

For $d > 4$ pad only the common summand: replace B by $B \times [0, 1]^{d-4}$ and embed K, L as the segments $[0, (b, 0)]$, $[0, (c, 0)]$. The three sums then acquire the same cube factor, so all four volumes above are unchanged, and $|K| = |L| = 0$ still; the same four integers refute the inequality in $\mathbb{R}^d$. Every quantity displayed in this proof is an integer or an exact rational, recomputed by `verify.py` in integer respectively rational arithmetic, and cross-checked in SageMath, which rebuilds each Minkowski sum as a vertex polytope over $\mathbb{Q}$ and computes its volume by triangulation instead. □

The author's own (non AI) proof. Unlike most of this archive, this refutation was first derived by hand, in May 2022 (independently of, and essentially contemporaneously with, the arXiv posting of Fradelizi et al. (2024b) in June 2022; no priority is claimed) by a three-step reduction that shows failure for every $d \ge 3$ without exhibiting a witness. It is the shortest disproof of Problem 1 we know of, so we record it in full. Throughout, a, b, c, h are convex bodies in $\mathbb{R}^d$, $V(a) = |a|$, and $V(a[d-k], h[k])$ is the mixed volume with a repeated $d - k$ times.

Step 1: drop a term. Applied to $(K, L, B) = (b, c, a)$ and stripped of the nonnegative term $|b|^{1/d}|c|^{1/d}$, the conjecture asserts $V(a)\,V(a + b + c) \le V(a + b)\,V(a + c)$, which rearranges to

$$f_b(a + c) \;\ge\; f_b(a) \qquad \text{for} \qquad f_b : \; a \;\longmapsto\; \frac{V(a)}{V(a + b)} :$$

for every fixed b, the functional f_b must be non-decreasing along Minkowski addition.

Step 2: linearize. The expansion $V(a + th) = \sum_{k \le d} \binom{d}{k} t^k\, V(a[d-k], h[k])$ is a polynomial in $t \ge 0$ with $\frac{d}{dt} V(a + th)\big|_{t=0} = d\, V(a[d-1], h)$, so $t \mapsto \log f_b(a + th)$ is differentiable; by Step 1 (applied at each base point $a + th$, with increment a multiple of h) it is non-decreasing, so its derivative at $t = 0$ is nonnegative:

$$\frac{V(a[d-1], h)}{V(a)} \;\ge\; \frac{V((a+b)[d-1], h)}{V(a + b)} \qquad \text{for all bodies } a, b, h.$$

Step 3: take h a ball. Since $d\, V(a[d-1], B_2^d) = |\partial a|$ is surface area, the display at $h = B_2^d$ is exactly the monotonicity $I(a + b) \ge I(a)$ of the information functional $I(\cdot) = |\cdot|/|\partial \cdot|$ of Dembo et al. (1991). Artstein-Avidan et al. (2014, Section 4.1) exhibit, for every $d \ge 3$, an interval b and a truncated box a with $I(a + b) < I(a)$ — and prove that monotonicity does hold for $d = 2$,

consistent with the planar case of the conjecture being true. Undoing Step 2, the strictly negative derivative gives a $t > 0$ with $V(a+b)\,V(a+tB_2^d) < V(a)\,V(a+b+tB_2^d)$, and the conjecture fails at $(K, L, B) = (b, tB_2^d, a)$.

The explicit zonotope witness above was constructed for this archive in August 2026, so that the case is self-contained and exactly recomputable; it lands in $d = 4$ of necessity, since log-submodularity of volume holds for zonoids in $d \le 3$ (Fradelizi et al., 2024a).

What holds instead: hyperbolic polynomials Replace $\mathrm{Vol}^{1/d}$ on convex bodies by $p^{1/m}$ on the closed hyperbolicity cone of a hyperbolic polynomial p of degree m. Then, the four-point inequality becomes a theorem, since Theorem A.4 in Appendix A proves

$$p^{1/m}(a+b)\,p^{1/m}(a+c) \;\ge\; p^{1/m}(a)\,p^{1/m}(a+b+c) + p^{1/m}(b)\,p^{1/m}(c),$$

extra term included. The weaker inequality without the $p^{1/m}(b)\,p^{1/m}(c)$ was first observed by the author (S. Sra) here (Sra, 2017).

Scope, and others' counterexamples. The conjecture has by now been refuted several times over. The sharp Plünnecke–Ruzsa constants of Fradelizi et al. (2024b) already imply that log-submodularity — hence the stronger inequality here — fails for general convex bodies in every dimension $d \ge 3$. In August 2026, Skorupinski (2026) independently gave a zonotope counterexample to log-submodularity in $\mathbb{R}^4$, with volume vector $(|B|, |K+B|, |L+B|, |K+L+B|) = (14, 30, 26, 56)$. The witness above came from an independent random search over integer generators, which turned up dozens of violating configurations — among them, rediscovered up to a unimodular transformation, Skorupinski's own witness with its volume vector $(14, 30, 26, 56)$ — so such violations are not isolated. The configuration kept here is a different one: its volume vector $(28, 68, 44, 108)$ is not proportional to his, so the two are not affinely equivalent, and its relative margin is about twice as large. What none of these witnesses touch is the variant of the question in which the common summand B is a Euclidean ball — the case closest to W Gaussian in the entropy-power inequality it mirrors, and one where the linearization above is provably monotone (Dembo et al., 1991).

Appendix A. Two new results for hyperbolic polynomials (author: S. Sra)

These results are non-AI (by S. Sra); including here to share with interested folks.

Replacing "Lorentzian" by "hyperbolic" turns the refuted principle into a theorem. Let $p : \mathbb{R}^n \to \mathbb{R}$ be homogeneous of degree $m \ge 1$ and hyperbolic with respect to e, normalized so that $p(e) > 0$, and let Λ_{++} be the open hyperbolicity cone containing e. By Gårding (1959), Λ_{++} is an open convex cone, p is hyperbolic with respect to each of its points, and $F := p^{1/m}$ is positive, homogeneous of degree one and concave on it. The other external input is the Helton–Vinnikov theorem (Helton and Vinnikov, 2007; Lewis et al., 2005), in the normalized form: a ternary form q of degree m, hyperbolic with respect to $\mathbf{1} = (1,1,1)$ with $q(\mathbf{1}) > 0$, admits real symmetric A_1, A_2, A_3 with $A_1 + A_2 + A_3 = I_m$ and $q(r) = q(\mathbf{1})\det(r_1A_1 + r_2A_2 + r_3A_3)$. Three points at a time is exactly what a triangle inequality needs.

Lemma A.1 (Ternary restriction)**.** *Fix $x_1, x_2, x_3 \in \Lambda_{++}$, put $T(r) := r_1x_1 + r_2x_2 + r_3x_3$ and $q := p \circ T$. Then q is homogeneous of degree m and hyperbolic with respect to $\mathbf{1}$, and each coordinate vector $\varepsilon_i \in \mathbb{R}^3$ lies in the hyperbolicity cone of q containing $\mathbf{1}$.*

Proof. Homogeneity is immediate, and $w := T(\mathbf{1}) = x_1+x_2+x_3 \in \Lambda_{++}$ because the cone is convex. So p is hyperbolic with respect to w, and for every $r \in \mathbb{R}^3$ the univariate $\lambda \mapsto q(r+\lambda\mathbf{1}) = p(T(r)+\lambda w)$ has only real zeros; that is hyperbolicity with respect to $\mathbf{1}$. For the second claim, the path $\gamma_i(t) = (1-t)\varepsilon_i + t\mathbf{1}$ satisfies $T(\gamma_i(t)) = x_i + t\sum_{j\ne i} x_j \in \Lambda_{++}$, so $q(\gamma_i(t)) > 0$ on $[0,1]$: it joins ε_i to $\mathbf{1}$ inside the positive component. □

Lemma A.2 (Positive definite representatives)**.** *In that setting the matrices A_1, A_2, A_3 supplied by Helton–Vinnikov are positive definite, and for all i, j,*

$$p(x_i) = q(\mathbf{1}) \det A_i, \qquad p\Big(\frac{x_i + x_j}{2}\Big) = q(\mathbf{1}) \det\Big(\frac{A_i + A_j}{2}\Big).$$

Proof. Along the path of Lemma A.1, $L_i(t) := \sum_k \gamma_i(t)_k A_k$ satisfies $q(\gamma_i(t)) = q(\mathbf{1}) \det L_i(t) > 0$, so $L_i(t)$ is nonsingular on $[0, 1]$; the eigenvalues of a continuous path of nonsingular real symmetric matrices cannot change sign, so the inertia is constant. At $t = 1$, $L_i(1) = A_1 + A_2 + A_3 = I_m \succ 0$, hence $A_i = L_i(0) \succ 0$. The two identities are the determinantal representation evaluated at ε_i and at $(\varepsilon_i + \varepsilon_j)/2$. □

Theorem A.3 ([**Proof**]: hyperbolic Jensen–Stein pseudometric)**.** *Let p be hyperbolic as above. Then $\sqrt{\mathsf{J}_p}$, with $\mathsf{J}_p(x, y) = \log p(\frac{x+y}{2}) - \frac{1}{2}\log p(x) - \frac{1}{2}\log p(y)$, is a pseudometric on Λ_{++}.*

Proof. Concavity of $F = p^{1/m}$ gives $F(\frac{x+y}{2}) \geq \frac{F(x)+F(y)}{2} \geq \sqrt{F(x)F(y)}$, which after raising to the mth power makes $\mathsf{J}_p \geq 0$; symmetry and vanishing on the diagonal are immediate. For the triangle inequality fix $x_1, x_2, x_3 \in \Lambda_{++}$ and take $A_1, A_2, A_3 \succ 0$ from Lemma A.2. The factor $q(\mathbf{1})$ cancels, leaving

$$\mathsf{J}_p(x_i, x_j) = \log \frac{\det((A_i + A_j)/2)}{\sqrt{\det A_i \det A_j}} = \delta_{\mathrm{S}}^2(A_i, A_j),$$

the squared Stein divergence, whose square root is a metric on the positive definite cone (Sra, 2016). Applying that triangle inequality to A_1, A_2, A_3 gives the claim, and the triple was arbitrary. □

Theorem A.4 below proves a conjecture once shared with the author (S. Sra) by Fraser (2025).

Theorem A.4 ([**Proof**])**.** *With $F = p^{1/m}$, for all a, b, c in the closed hyperbolicity cone,*

$$F(a + b)\, F(a + c) \;\geq\; F(a)\, F(a + b + c) + F(b)\, F(c).$$

Proof. Write $\mathcal{D}(M) = \det(M)^{1/m}$. We first prove the matrix form $\mathcal{D}(A+B)\mathcal{D}(A+C) \geq \mathcal{D}(A)\mathcal{D}(A+B+C) + \mathcal{D}(B)\mathcal{D}(C)$ for $A, B, C \succeq 0$. For $A \succ 0$, put $X = A^{-1/2}BA^{-1/2}$ and $Y = A^{-1/2}CA^{-1/2}$; every term carries $\mathcal{D}(A)^2$, so it suffices to show $\mathcal{D}(I + X)\mathcal{D}(I + Y) \geq \mathcal{D}(I + X + Y) + \mathcal{D}(X)\mathcal{D}(Y)$. Order the eigenvalues of X decreasingly and those of Y increasingly. Since a product does not depend on the ordering, $\mathcal{D}(I + X)\mathcal{D}(I + Y) = \big(\prod_i[(1 + x_i + y_i) + x_i y_i]\big)^{1/m}$, and Minkowski's determinant inequality $\det(R + S)^{1/m} \geq \det R^{1/m} + \det S^{1/m}$ — itself Jensen's inequality for $t \mapsto \log(1 + e^t)$ applied to the eigenvalues of $R^{-1/2}SR^{-1/2}$ — applied to the diagonal matrices $R = \mathrm{diag}(1 + x_i + y_i)$ and $S = \mathrm{diag}(x_i y_i)$ gives

$$\mathcal{D}(I + X)\mathcal{D}(I + Y) \;\geq\; \Big(\prod_i (1 + x_i + y_i)\Big)^{1/m} + \Big(\prod_i x_i y_i\Big)^{1/m}.$$

Fiedler's upper bound (Fiedler, 1971), which pairs one decreasing spectrum with the other increasing one, gives $\det(I + X + Y) \leq \prod_i (1 + x_i + y_i)$, while the second term is $\mathcal{D}(X)\mathcal{D}(Y)$; a limit $\varepsilon \downarrow 0$ in $A + \varepsilon I$ covers singular A. For the cone statement, apply Lemmas A.1 and A.2 to $a, b, c \in \Lambda_{++}$ to get $A, B, C \succ 0$ with $p(ra + sb + tc) = q(\mathbf{1}) \det(rA + sB + tC)$; then $F(ra + sb + tc) = q(\mathbf{1})^{1/m}\mathcal{D}(rA + sB + tC)$ for $r, s, t \geq 0$, and the displayed inequality is the matrix one times $q(\mathbf{1})^{2/m}$. Points of the closed cone are handled by $u \mapsto u + \varepsilon e$, which lies in Λ_{++}, and continuity. □

Nothing here collides with Theorem 4 and Corollary 5, and it is worth saying why not. The witness (1) is strictly Lorentzian but *not* hyperbolic: $G(x, 1) = \frac{1}{10}x^3 + \frac{3}{2}x^2 + \frac{11}{5}x + 1$ has discriminant

$$18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2 = -\frac{1499}{1250} < 0 \qquad \Big(a, b, c, d = \tfrac{1}{10}, \tfrac{3}{2}, \tfrac{11}{5}, 1\Big),$$

so it has one real and two complex conjugate roots. A binary form is hyperbolic with respect to some direction only if it splits into real linear factors — each factor ℓ_i contributes the root $-\ell_i(x)/\ell_i(e)$,

which is real for every real x only when ℓ_i is real up to scale — so G is hyperbolic with respect to no direction at all. In two variables hyperbolicity is therefore strictly stronger than strict Lorentzianity, and the witness lives in the gap. That gap is the content of both results together: what makes the midpoint Jensen gap a squared distance is the determinantal representation that hyperbolicity buys through Helton–Vinnikov, and not log-concavity of the coefficients.

Code and certificate

The case is the directory `counterexamples/courtade-volume-conjecture/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. Independent Sage interval cross-checks: `verify_courtade.sage`. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/courtade-volume-conjecture
python verify.py
```

- Theorem 2 (result `courtade-volume-conjecture`): exact arithmetic.
- `artifacts/certificate.json`: Integer generators of the zonotope and the two segments, the four integer volumes 28, 68, 44, 108, the violated products $2992 < 3024$, and the exact rational volumes of the thickened full-dimensional variant

AI usage

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by Suvrit Sra, by hand (2022-05). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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